

24/24

Excellent!

100%

Name: _____

These questions assess your understanding of the material from Chapter(s) 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Suppose a galaxy is moving away from Earth at a speed of 2.6×10^8 m/s. It emits radio waves with a wavelength of 0.37 m. What wavelength would we detect on Earth?

ANSWER: 1.38 m ✓

$$\lambda_{cb} = \lambda_s \sqrt{\frac{1+u/c}{1-u/c}}$$

$$\rightarrow 0.37(3.742)$$

$$\sqrt{\frac{1+u/c}{1-u/c}} \rightarrow \frac{u}{c} = \frac{2.6 \times 10^8}{3 \times 10^8} = 0.867$$

$$\sqrt{\frac{1.867}{0.1333}} = 3.742$$

2. What angle would the axis of a polarizing filter need to make with the direction of polarized light of intensity 1.9 kW/m^2 to reduce the intensity to 0.60 kW/m^2 ?

ANSWER: 55.81° ✓

$$I = I_0 \cos^2 \theta$$

$$\rightarrow \cos \theta = \sqrt{\frac{I}{I_0}} \rightarrow \sqrt{\frac{0.6}{1.9}} = 0.562$$

$$\theta = \cos^{-1}(0.562)$$

3. A scientist at a natural-history museum is holding a specimen 29. cm from a thin convex magnifying glass of focal length 20. cm. What magnification is produced?

ANSWER: -2.22 ✓

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\rightarrow \frac{1}{d_i} = \frac{1}{0.2} - \frac{1}{0.29}$$

$$d_i = 0.644$$

$$m = -\frac{d_i}{d_o}$$

$$\rightarrow -\frac{0.644}{0.29} + 3$$

- ① 4. Calculate the minimum thickness of an oil slick on water that appears blue when illuminated by white light perpendicular to its surface. Take the blue wavelength to be 450 nm, $n_{\text{oil}} = 1.49$ and $n_{\text{water}} = 1.31$.

$$2nt = (m + \frac{1}{2})\lambda \quad m=0$$

ANSWER: $7.55 \times 10^{-8} \text{ m}$

$$\rightarrow t = \frac{\lambda}{4n_{\text{oil}}}$$

$$\rightarrow \frac{450 \times 10^{-9}}{(1.49)^4}$$

- ✓ 5. A medical imaging device produces ultrasound by oscillating with a period of $0.78 \mu\text{s}$. What is the frequency of this oscillation?

ANSWER: $1.28 \times 10^6 \text{ Hz}$

$$f = \frac{1}{T}$$

$$\rightarrow \frac{1}{0.78 \times 10^{-6}}$$

- ✓ 6. Calculate the motional emf induced along a 20.-km-long conducting object moving at an orbital speed of $10. \frac{\text{km}}{\text{s}}$ perpendicular to Earth's 63.- μT magnetic field.

ANSWER: $1.26 \times 10^4 \text{ V}$

$$\mathcal{E} = Blv$$

$$\rightarrow (63 \times 10^{-6})(20 \times 10^3)(10 \times 10^3)$$

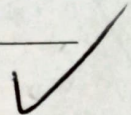
+2

7. How much energy is stored in a 2.8-mH inductor when a 13.-A current flows through it?

ANSWER: 0.245

$$E_{\text{ind}} = \frac{1}{2} L I^2$$

$$\rightarrow \frac{1}{2} (2.8 \times 10^{-3}) 13^2$$

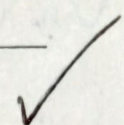


8. What is the resistance of an automobile headlight which carries a current of 1.3 A when 5.2 V is applied to it?

ANSWER: 4 Ω

$$V = IR \rightarrow R = \frac{V}{I}$$

$$\rightarrow \frac{5.2}{1.3}$$



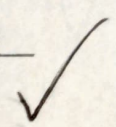
9. A car headlight filament is made of tungsten ($\rho = 5.6 \times 10^{-8} \Omega \cdot \text{m}$) and has a resistance of 0.39 Ω. If the filament is a cylinder 23.-cm long, what is its diameter?

ANSWER: $2.05 \times 10^{-4} \text{ m}$

$$R = \frac{\rho L}{A} \rightarrow A = \frac{\rho L}{R}$$

$$\rightarrow \frac{(5.6 \times 10^{-8}) 0.23}{0.39} = 3.303 \times 10^{-8}$$

$$A = \pi r^2 \rightarrow d = \sqrt{\frac{4A}{\pi}} \rightarrow \sqrt{\frac{4(3.303 \times 10^{-8})}{\pi}}$$



+ 3

$$d = \sqrt{\frac{4A}{\pi}}$$

$$A = \pi r^2$$

- ✓ 10. Calculate the energy that is released upon the complete annihilation of a 4.1-g object.
 $E_0 = mc^2$
 ANSWER: $3.69 \times 10^{14} \text{ J}$

$$\rightarrow (4.1 \times 10^{-3}) (3 \times 10^8)^2$$

- ✓ 11. An electron moves at $1.1 \times 10^7 \frac{\text{m}}{\text{s}}$ perpendicular to a magnetic field. The field causes the electron to travel in a circular path of radius 47. cm. What is the strength of the magnetic field?

$$r = \frac{mv}{qB} \rightarrow B = \frac{mv}{rq}$$

ANSWER: $1.33 \times 10^{-4} \text{ T}$

$$\rightarrow \frac{(9.11 \times 10^{-31}) (1.1 \times 10^7)}{0.47 (1.6 \times 10^{-19})}$$

- ✓ 12. A wire carries a 49. A current toward the south. If a large magnet were placed directly to the west of the wire and emits a 0.82 T magnetic field perpendicularly towards the wire, calculate the force exerted on the wire per length of wire.

$$F = I l B \sin \theta \quad \theta = 90^\circ$$

ANSWER: 40.2 N/m

$$\rightarrow \frac{F}{l} = IB \rightarrow 49(0.82)$$

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13. Calculate the magnitude of the force exerted on a 0.10-nC charged particle placed in a $1.5 \times 10^4 \frac{\text{N}}{\text{C}}$ electric field.

$$E = \frac{F}{q}$$

ANSWER: $1.5 \times 10^{-6} \text{ N}$ ✓

$$\Rightarrow F = E q \Rightarrow (1.5 \times 10^4) (0.1 \times 10^{-9})$$

14. An incandescent lightbulb has a 0.79-cm-long filament and operates at 115. V to deliver 65. W. Calculate the diameter of the bulb's filament, assuming it to be a cylindrical tungsten wire, given that the resistivity of tungsten is $5.6 \times 10^{-8} \Omega \cdot \text{m}$.

ANSWER: _____

(2)

15. Calculate the magnitude of an electric field measured at the origin due to the presence of two point charges q_1 , q_2 . q_1 is located 0.14 cm south of the origin and q_2 is located 0.35 cm east of the origin. $q_1 = -8.8 \text{ nC}$ and $q_2 = +1.5 \text{ nC}$.

ANSWER: $4.64 \times 10^7 \text{ N/C}$ ✓

$$E = k \frac{|q|}{r^2}$$

$$\Rightarrow E_1 = \frac{(8.99 \times 10^9) (-8.8 \times 10^{-9})}{0.0014^2} = -4.04 \times 10^7$$

$$\Rightarrow E_2 = \frac{(8.99 \times 10^9) (1.5 \times 10^{-9})}{0.0035^2} = 1.101 \times 10^6$$

$$\sqrt{E_1^2 + E_2^2}$$

$$\Rightarrow \sqrt{(-4.04 \times 10^7)^2 + (1.101 \times 10^6)^2}$$

+2

- ✓ 16. A sonar echo returns to a submarine 1.99 s after being emitted from the submarine. What is the distance from the submarine to the object, given that the speed of sound in this water is $1.49 \frac{\text{km}}{\text{s}}$?

ANSWER: $1.48 \times 10^3 \text{ m}$

$$v = \frac{d}{t} \rightarrow d = vt$$

$$\rightarrow \frac{(1.49 \times 10^3) 1.99}{2}$$

- ✓ 17. Calculate the power of the eye when viewing an object 6.4 cm away. Take the lens-to-retina distance to be 2.6 cm.

ANSWER: 54.1 P

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = \frac{1}{f}$$

$$\rightarrow \frac{1}{0.004} + \frac{1}{0.026}$$

- ✓ 18. Suppose the voltage supplied to a circuit is 11. V. There are three resistors wired in parallel: $R_1 = 4.4 \Omega$, $R_2 = 6.1 \Omega$, and $R_3 = 5.1 \Omega$. Calculate the power output of the battery.

ANSWER: 71.06 W

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \rightarrow \frac{1}{4.4} + \frac{1}{6.1} + \frac{1}{5.1}$$

$$R_p = 1.703$$

$$P = \frac{V^2}{R} \rightarrow \frac{11^2}{1.703}$$

+3

19. Suppose a train sounding its 121.-Hz horn is moving at $23.6 \frac{m}{s}$. What frequency does a person standing by the tracks hear after the train passes? Given this particular day's temperature, the speed of sound is $328. \frac{m}{s}$.

ANSWER: 112.88 Hz ✓

$$f_{obs} = f_s \left(\frac{v_w}{v_w \pm v_s} \right)$$

$$\rightarrow 121 \left(\frac{328}{328 + 23.6} \right)$$

$$v_w = 331 \checkmark$$

20. What is the electric potential 0.41 cm away from the center of a metal sphere that has a 3.3-nC static charge?

ANSWER: $7.24 \times 10^5 V$ ✓

$$V = \frac{kQ}{r}$$

$$\rightarrow \frac{(8.99 \times 10^9) (3.3 \times 10^{-9})}{0.0041}$$

21. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the period of oscillations for a car with shoddy shock absorbers if the car's total mass is 805. kg and the force constant of the suspension system is $3.23 \times 10^4 \frac{N}{m}$.

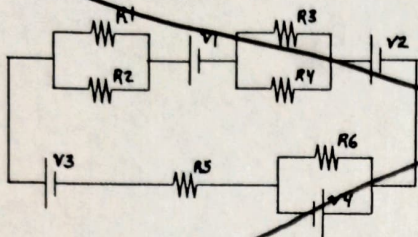
ANSWER: 0.992 s ✓

$$T_{SHO} = 2\pi \sqrt{\frac{m}{k}}$$

$$\rightarrow 2\pi \sqrt{\frac{805}{3.23 \times 10^4}}$$

22. Solve for the voltage drop across resistor R_6 given $V_1 = 10\text{ V}$, $V_2 = 8\text{ V}$, $V_3 = 3\text{ V}$, $V_4 = 4\text{ V}$, $R_1 = 8\ \Omega$, $R_2 = 8\ \Omega$, $R_3 = 2\ \Omega$, $R_4 = 2\ \Omega$, $R_5 = 1\ \Omega$, and $R_6 = 3\ \Omega$.

ANSWER: _____



23. What power of spectacle lens is needed to correct the vision of a near-sighted person whose far point is 160 cm? Assume the spectacle lens is held 1.3 cm away from the eye by eyeglass frames.

ANSWER: -0.63 D

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$d_o = \infty$$

$$\rightarrow \frac{1}{f} = \frac{1}{d_i} = P$$

$$\rightarrow \frac{1}{-1.587}$$

$$d_i = -(1.6 - 0.013)$$

24. Calculate the sound intensity level in decibels for a sound wave traveling in air at 2.1°C with a pressure amplitude of 0.722 Pa .

Recall that the density of air is $1.29 \frac{\text{kg}}{\text{m}^3}$ and the speed of sound in air at 0°C is $331 \frac{\text{m}}{\text{s}}$.

$$\beta(\text{dB}) = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

ANSWER: 87.84 dB

$$I_0 = 1 \times 10^{-12}$$

$$I = \frac{P^2}{2\rho v_w} \rightarrow \frac{0.722^2}{2(1.29)332.271} = 6.081 \times 10^{-4}$$

$$v_w = 331 \sqrt{\frac{T}{273}} \rightarrow 331 \sqrt{\frac{275.1}{273}} = 332.271$$

+1

$$\rightarrow 10 \log_{10} \left(\frac{6.081 \times 10^{-4}}{1 \times 10^{-12}} \right)$$

25. On its highest power setting, a 730-W microwave oven projects microwaves onto a 0.45-m-by-0.45-m area. Calculate the peak electric-field strength in the electromagnetic waves emitted by the microwave oven.

ANSWER: _____

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26. Suppose that you would like to find the angle of refraction in a material. You shine a laser through air at an incident angle of 9.6° into the material which has an index of refraction of 1.5. Calculate the angle of refraction in the material.

ANSWER: 6.38° ✓

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\rightarrow \sin \theta_2 = \frac{n_1 \sin \theta_1}{n_2} \rightarrow \frac{\sin 9.6^\circ}{1.5} = 0.111$$

$$\theta_2 = \sin^{-1}(0.111)$$

27. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of 1.5×10^3 V/m?

ANSWER: 5×10^{-6} T ✓

$$c = \frac{E}{B} \rightarrow B = \frac{E}{c}$$

$$\rightarrow \frac{1.5 \times 10^3}{3 \times 10^8}$$

+ 02

- ✓ 28. Find the total capacitance for three capacitors connected in series, given that their individual capacitances are 1.9 nF, 14. nF, and 26. nF.

ANSWER: $1.572 \times 10^{-9} \text{ F}$

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\rightarrow \frac{1}{(1.9 \times 10^{-9})} + \frac{1}{(14 \times 10^{-9})} + \frac{1}{(26 \times 10^{-9})} = \frac{1}{6.362 \times 10^{-8}}$$

- ✓ 29. Calculate the critical angle for light traveling in a material that is surrounded by air, given that the material has an index of refraction of 1.6.

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad \theta = 90$$

ANSWER: 38.68°

$$\rightarrow \sin \theta_c = \frac{n_2 \sin \theta_2}{n_1} \rightarrow \frac{1}{1.6}$$

$$\theta_c = \sin^{-1}(0.625)$$

30. Calculate the radius of curvature of the path of a proton having a velocity of $1.1 \times 10^7 \frac{\text{m}}{\text{s}}$ perpendicular to a magnetic field of strength $B = 0.13 \text{ T}$.

ANSWER: _____

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1. $\lambda_{obs} = 1.38 \text{ m} \therefore \lambda_{obs} = \lambda_s \sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}$
(Chapter 28, Section (4) Relativistic Addition of Velocities - cple_ch28_ex5)
2. $\theta = 55.8^\circ \therefore I = I_0 \cos^2 \theta$
(Chapter 27, Section (8) Polarization - cple_ch27_p87)
3. $m = -2.22 \therefore \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$
(Chapter 26, Section (6) Image Formation by Lenses - cple_ch25_ex7)
4. $t = 7.55 \times 10^{-8} \text{ m} \therefore 2t = \left(m + \frac{1}{2}\right) \lambda_n, n_{air} < n_{oil} > n_{water}$
(Chapter 27, Section (7) Thin-Film Interference - cple_ch27_p72)
5. $f = 1.28 \times 10^6 \text{ Hz} \therefore T = \frac{1}{f}$
(Chapter 16, Section (2) Period and Frequency in Oscillations - cple_ch16_ex3)
6. $\varepsilon = 1.26 \times 10^4 \text{ V} \therefore \varepsilon = B\ell v$
(Chapter 23, Section (3) Motional EMF - cple_ch23_ex2)
7. $E_{ind} = 0.237 \text{ J} \therefore E_{ind} = \frac{1}{2} LI^2$
(Chapter 23, Section (9) Inductance - cple_ch23_ex8)
8. $R = 4.00 \Omega \therefore V = IR$
(Chapter 20, Section (2) Ohm's Law - cple_ch20_ex4)
9. $D = 2.05 \times 10^{-4} \text{ m} \therefore R = \frac{\rho L}{A}$
(Chapter 20, Section (3) Resistance and Resistivity - cple_ch20_ex5)
10. $E = 3.69 \times 10^{14} \text{ J} \therefore E_0 = mc^2$
(Chapter 28, Section (6) Relativistic Energy - cple_ch28_ex6)
11. $B = 1.33 \times 10^{-4} \text{ T} \therefore r = \frac{mv}{qB}, m_e = 9.11 \times 10^{-31} \text{ kg}, q_e = 1.60 \times 10^{-19} \text{ C}$
(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cple_ch22_p13)
12. $\frac{F}{\ell} = 40.2 \frac{\text{N}}{\text{m}} \therefore F = I\ell B \sin \theta, \theta = 90^\circ$
(Chapter 22, Section (7) Magnetic Force on a Current-Carrying Conductor - cple_ch22_ex4)
13. $F = 1.50 \times 10^{-6} \text{ N} \therefore F = Eq$
(Chapter 18, Section (4) Electric Field - cple_ch18_ex3)
14. $d = 1.66 \times 10^{-6} \text{ m} \therefore P = \frac{V^2}{R}, R = \frac{\rho L}{A} \text{ and } A = \pi r^2$
(Chapter 20, Section (4) Electric Power and Energy - tkd2)
15. $E = 4.04 \times 10^7 \frac{\text{N}}{\text{C}} \therefore E = k \frac{|Q|}{r^2}, E^2 = E_x^2 + E_y^2$
(Chapter 18, Section (5) Electric Field Lines - cple_ch18_ex4)
16. $d = 1.48 \times 10^3 \text{ m} \therefore \vec{v}_{avg} = \frac{\Delta \vec{x}}{\Delta t}$
(Chapter 17, Section (2) Speed of Sound, Frequency, Wavelength - cple_ch17_p8)

$$17. P = 54.1 \text{ D} \because \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$$

(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex2b)

$$18. P = 71.1 \text{ W} \because \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}, V = IR, P = IV$$

(Chapter 21, Section (1) Resistors in Series and Parallel - cp1e_ch21_ex2)

$$19. f = 112.9 \text{ Hz} \because f_{obs} = f_s \left(\frac{v_w}{v_w \pm v_s} \right) \text{ (review the Doppler effect)}$$

(Chapter 17, Section (4) Doppler Effect and Sonic Booms - cp1e_ch17_ex4)

$$20. \Delta V = 7.24 \times 10^3 \text{ V} \because V = \frac{kQ}{r}$$

(Chapter 19, Section (3) Electric Potential Due to a Point Charge - cp1e_ch19_ex6)

$$21. f = 0.992 \text{ s} \because T_{SHO} = 2\pi \sqrt{\frac{m}{k}}$$

(Chapter 16, Section (3) Simple Harmonic Motion - cp1e_ch16_ex4)

$$22. V = 4.00 \text{ V} \because \text{Kirchhoff's rules and } V = IR$$

(Chapter 21, Section (3) Kirchhoff's Rules - tkd_circ2)

$$23. P = -0.630 \text{ D} \because \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$$

(Chapter 26, Section (2) Vision Correction - cp1e_ch26_ex3)

$$24. \beta = 87.8 \text{ dB} \because v_w = \left(331 \frac{\text{m}}{\text{s}} \right) \sqrt{\frac{T}{273 \text{ K}}}, I = \frac{(\Delta p)^2}{2\rho v_w}, \beta \text{ (dB)} = 10 \log_{10} \left(\frac{I}{I_0} \right), I_0 = 1 \times 10^{-12} \frac{\text{W}}{\text{m}^2}$$

(Chapter 17, Section (3) Sound Intensity and Sound Level - cp1e_ch17_ex2)

$$25. E_0 = 1.65 \times 10^3 \frac{\text{V}}{\text{m}} \because I_{ave} = \frac{c\epsilon_0 E_0^2}{2} = \frac{cB_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$$

(Chapter 24, Section (4) Energy in Electromagnetic Waves - cp1e_ch24_ex4)

$$26. \theta_2 = 6.38^\circ \because n_1 \sin \theta_1 = n_2 \sin \theta_2$$

(Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex3)

$$27. B_{max} = 5.00 \times 10^{-6} \text{ T} \because c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E}{B} = f\lambda$$

(Chapter 24, Section (2) Production of Electromagnetic Waves - cp1e_ch24_ex1)

$$28. C_T = 1.57 \times 10^{-9} \text{ F} \because \frac{1}{C_s} = \frac{1}{C_1} + \dots + \frac{1}{C_n}$$

(Chapter 19, Section (6) Capacitors in Series and Parallel - cp1e_ch19_ex9)

$$29. \theta_c = 38.7^\circ \because n_1 \sin \theta_1 = n_2 \sin \theta_2$$

(Chapter 25, Section (4) Total Internal Reflection - cp1e_ch25_ex4)

$$30. r = 0.883 \text{ m} \because r = \frac{mv}{qB}, m_p = 1.67 \times 10^{-27} \text{ kg}, q_p = 1.60 \times 10^{-19} \text{ C}$$

(Chapter 22, Section (5) Force on a Moving Charge in a Magnetic Field - cp1e_ch22_ex2)